

Topic: Optical Imaging and Deep Learning Framework for Real-Time Detection of Bacterial Pathogens in Water

Detecting harmful bacteria in drinking water remains a critical public health challenge. Although indicator organisms such as *Escherichia coli* and fecal streptococci are widely used to indicate faecal contamination, conventional detection methods are costly, time-consuming, and require specialized laboratories and personnel. This study introduces an optical water-droplet imaging approach combined with deep learning, which enables the rapid, accurate, and cost-effective detection and classification of these bacterial indicators. First, a baseline convolutional neural network (CNN) was developed to classify optically acquired droplet images. This network achieved a classification accuracy of 89% with a loss of 0.13. On this foundation, transfer learning was applied using pre-trained architectures, including ResNet-50, ResNet-152, DenseNet-201, and EfficientNet-B7, to distinguish four classes: *E. coli*, fecal streptococci, the co-occurrence of both, and safe water. The improved framework achieved significantly better performance. ResNet-50, ResNet-152, and EfficientNet-B7 reached accuracies of 94.15%, 94.45%, and 95.54%, respectively. Corresponding precision, recall, and F1 scores exceeded 94%. Receiver operating characteristic (ROC) analysis further confirmed strong discriminative capability, with an area under the curve (AUC) above 98% for each bacterial class. These results demonstrate the feasibility of an integrated, real-time, optical-based detection system for waterborne pathogens. This system offers a scalable, automated solution that aligns with sustainable water, sanitation, and hygiene (WASH) objectives.

Biography: I am pursuing a Ph.D. in Physics at the University of Dschang, where my research focuses on the optical detection and classification of waterborne disease agents to support safer water quality monitoring. I also serve as a teaching assistant at the African Institute for Mathematical Sciences (AIMS), Cameroon. Ultimately, I aim to apply deep learning, image processing, and computer vision to detect and classify microorganisms in water, food, and bodily fluids, addressing real-world public health challenges.